Mechanically Alloyed Magnesium Based Nanostructured Alloy Powders for Biomedical Applications

P. Morcos, K.I. ElKhodary, and H.G. Salem

Abstract

Mechanical alloying (MA) is one of the most commonly used methods in preparing metallic materials. We are especially interested in its ability to produce nanostructured materials. MA has rarely been used in producing nanostructured magnesium base alloys due to the process complexity and flammable nature of the alloy. In this work, high energy ball milling of elemental powders, Mg, Zn and Zr, was investigated for the MA of ZK50 alloy. Crystallite size, lattice strain, as a function of total milling energy (required for complete alloying) were characterized using X-ray diffraction. Scanning electron microscopy was employed to determine the influence of milling time on powder morphology. Consolidation of the milled powders into thin sheets was conducted via hot pipe rolling at 0.4 T_m. Results showed that complete alloying was achieved at 45 milling hours which was associated with particle size reduction to 600 nm and crystallite size of 8.8 nm.

Keywords

Magnesium ZK50 • Mechanical alloying • Stents

Introduction

Materials used for biomedical applications such as orthopedic and cardiac stents can be classified into three main categories; polymers, ceramics and metals. For load bearing applications, metal based implants such as stainless steel, Ti-based alloys and Co-Cr based alloys, are considered the best for their higher strength and stiffness compared to polymers and ceramics [1]. However, they have high mismatch between their properties and the properties of the human bones due to the difference in the mechanical properties between them. Magnesium has mechanical properties similar to human bones. Its density is 1.738 gm/cm^3 while the density of human bone is 1.8–2.1 gm/cm^3 which is slightly higher. Besides, the elastic modulus of elemental Mg is 45 GPa, which fall within the human bone elastic modulus range (40–57 GPa). This gives advantage to Mg-based alloys, and prevents the stress shielding effect [2].

Mg is considered a biodegradable and biocompatible material, therefore, addition of some non-toxic elements to Mg can be used to produce alloys for cardiac stents with desired mechanical properties. Moreover, Mg and its alloys degrade in the human body with controlled rates and without causing any side effects to human organs [3]. The elements used for alloying of Mg are always chosen carefully for biomedical applications to prevent the formation of toxic compounds, which are deleterious to human body [4]. Besides, the amount of each element in the biodegradable alloy should not exceed a specific limit, otherwise it may cause toxicity to humans as well. Mg-based alloys can be classified into three main categories, Pure Mg with traces of alloying elements, Mg alloys with Aluminum (Al) main alloying element, and Al free Mg-based alloys. Al is very severe to the human body as it can cause nerve toxicity,

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Alzheimer's disease and restrain growth of the human body. Therefore, Al is not preferred as an alloying element for biodegradable materials [4]. On the other hand, Zinc (Zn) alloying is preferred, as it is present in the human body in low concentrations, and is less toxic. It is also very important for cell division and the synthesis of DNA and protein and essential for immune system [2]. The concentration of Zn in any alloy used for biomedical applications usually doesn't exceed 6% by weight [5].

Accordingly, selection of the alloying elements in Mg is based on the toxicity of the elements and the desired properties needed for the applications. Manganese (Mn), Calcium (Ca) and Zn are the essential elements for human body, they are typically first chosen as main alloying elements. Zn content up to 6% can be used in the alloy to release a safe amount daily for the body, besides, it enhances the ductility and the tensile strength of the alloy. This percentage is the highest amongst the three elements; Mn, Ca and Zn. The second alloying element is normally less in its mass percent (sometimes less than 1%) than the primary one; however, some elements can be added in very small percentage and cause a change in the mechanical properties of the whole alloy. Examples of those elements are Lithium (Li) and Zirconium (Zr) [5].

Ball Milling (BM) is considered a very powerful processing method for synthesizing of extremely fine microstructures [6]. Mg powders with or without other elements were investigated using BM by many authors for various applications. S. Hwang et al. appear to be the first group to study the mechanical milling of pure Mg powders [7]. Grosjean et al. also used BM to produced Mg powders with nanocrystallite size [8]. Other authors studied the preparation of Mg-Zn and Mg-Zr binary alloys. Palacios-Lazcano et al., used BM for preparing nanocrystalline structured alloys with 5 and 10 wt% Zn for hydrogen storage applications. They formed the alloy between 30 and 60 h of milling. The final alloy had a grain size less than 20 nm after 60 h [9]. Liu Zi-li et al., used BM to prepare Mg-1.5Zr alloys for damping applications. Other papers discussed BM of MgH [10], Mg-3Al-Zn [11], Mg-Ca [12] and Mg-Ti [6, 13].

In this paper, we introduce the BM process for alloying of Mg, Zn and Zr powders in order to produce ZK50 (Z refers to Zinc while K refers to Zirconium and the number represents the composition of each component) alloy which is new to the literature. In this alloy, Zn is chosen as it is one of the essential body elements while Zr is chosen because it is excellent for grain refinement, and will help obtain nano-sized grains. Moreover, Zr improves the mechanical properties, degradability, and has no effect on toxicity to humans. Furthermore, we introduce the powder rolling technique, which is novel to solid Mg based alloys sheets.

Materials and Methods

Elemental powders of Mg (purity >99.5%, Sigma Aldrich), Zn (purity >99.9%, Alfa Aesar), Zr (purity >99%, SE-JONG Materials) with average particle size of 44, 130, 44 μm, respectively were used as raw materials. A mixture of the powders was prepared under argon atmosphere in a glove box with nominal composition of Mg-5Zn-0.5Zr (mass fraction, %). The powders were then mixed in a turbula mixer at speed of 96 rpm for 60 min. The mixed powder was then milled in high-energy planetary ball milling (BM) machine; Retcsh PM400 for 45 h. 20 gm of the powders were placed in two stainless steel jars with stainless steel ball with 15:1 ball-to-powder ratio (BPR). Ethanol was added as Process Control Agent (PCA) during the process. Special cooling process in air and using refrigerator was applied for the jars at constant milling intervals to cool the powders to prevent powders' ignition.

The milled powders were canned in pure copper pipe 65 mm in length and 6 mm in inner diameter. Perfect sealing of the canned powders under argon atmosphere using a special locking tools and bench vise was done in order to prevent powders ignition at when subjected to high temperature. The pipes were hot rolled to obtain solid sheets of the alloy. Many trials different temperatures were attempted to produce well-consolidated thin solid sheets. Sound consolidated thin sheets were obtained via combination of hot pipe rolling at 300 °C with reheating intervals of 30 min. Details of the consolidation conditions are shown in Table 1.

Evaluation of the processed alloy sheets was measured using a Mettler Toledo XS 205 digital densitometer. The relative density (%) was calculated based on a ratio between the theoretical alloy density 1.813 gm/cc and the experimentally measured value. XRD Philips XPERT Powder Diffractometer with Cu K-Alpha Wavelength λ_{Cu} = 1.54056 Å was used to investigate the formation time of the alloy, track the reduction of crystallite size as well as the change in the lattice strain as a function of milling time. The lattice strain and crystallite size were determined using Sherrer Equation and Williamson-Hall Equations respectively. Post the refinement of the I-2θ peaks from noise. The average particle size of the milled powders was investigated using Leo55 Field Emission Scanning Electron Microscope (FE-SEM) with 1 nm resolution and 4 kV.

The consolidated sheets' hardness and Young's modulus were determined using a MTS Nanoindenter XP with a diamond Berkovich 50 nm indenter. At least 10 indentations (spaced 130 μm apart) were used to estimate the modulus and hardness of the specimen. The test was carried on the cross section of each specimen using Continuous Stiffness Measurement (CSM) Basic hardness and young's modulus

Table 1 Pipe rolling conditions: consolidated sheets of mechanically alloyed Mg-ZK50

Condition No.	Total % reduction	Number of passes	Sheet mean thickness (μm)	Process annealing temp/time
1	67	11	800	300 °C/30 min
2	72	12	720	300 °C/60 min

displacement control testing method with 2000 nm displacement into the surface.

Results and Discussion

Powders' Characterization

SEM images for the elemental powders are shown in Fig. 1. The Mg particles have a flake like morphology with an average particle size of 44 μm, purchased from Sigma Aldrich (Fig. 1a). Elemental Zn, which was purchased from Alfa Aesar are ellipsoidal and elongated particles about 210 μm in length and 50 μm in width as shown in Fig. 1b. Zr was received in the form of almost equiaxed particles from JONG Materials, with average particle size of 44 μm (Fig. 1c).

Figure 2a–g shows SEM images for the BM premixed powders as a function of milling time. It is clear from the displayed images that at early stages of milling of 3-h the individual Mg, Zn and Zr particles disappeared, which indicates the cold welding of the three elements. Cold welding dominated powder behavior for up to 5 h of milling (Fig. 2a, b). High-energy milling then resulted in the refinement of the powders, after 10-h of milling (Fig. 2c) followed by more refinement and clustering of the refined particles after 45-h of milling. Agglomeration and clustering of the refined particles dominated powder behavior at longer milling times, due to the high surface area associated with the refined microstructure. Similar observations were made by Grosjean et al. [8] while milling pure Mg powders for 40 h. Figure 3 shows a graph representing effect of milling time on the average particle size of the powders. The average particle size of the powders reached 600 nm after 45 h. It is clear from the displayed results in Fig, 3 that most of the particle size reduction occurred during 19-h of milling as the particle size reach to 1.3 μm followed by gradual reduction down to 600 nm by the end of the milling process.

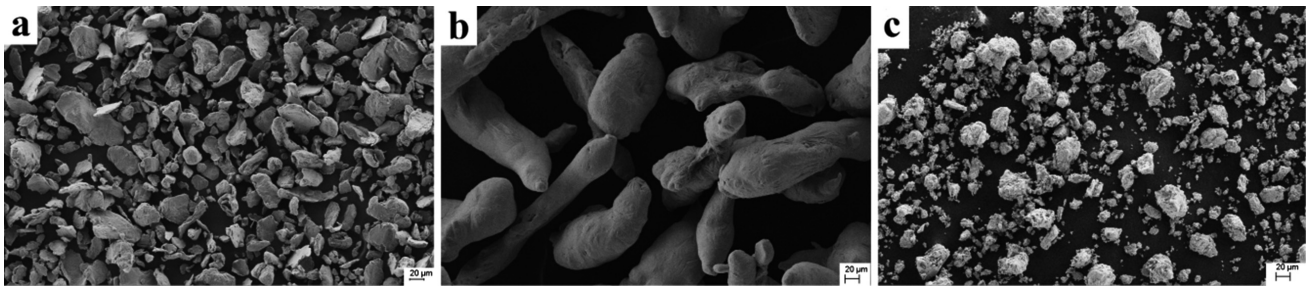


Fig. 1 SEM images for the as-received elemental powders at 500× **a** Magnesium, **b** Zinc and **c** Zirconium

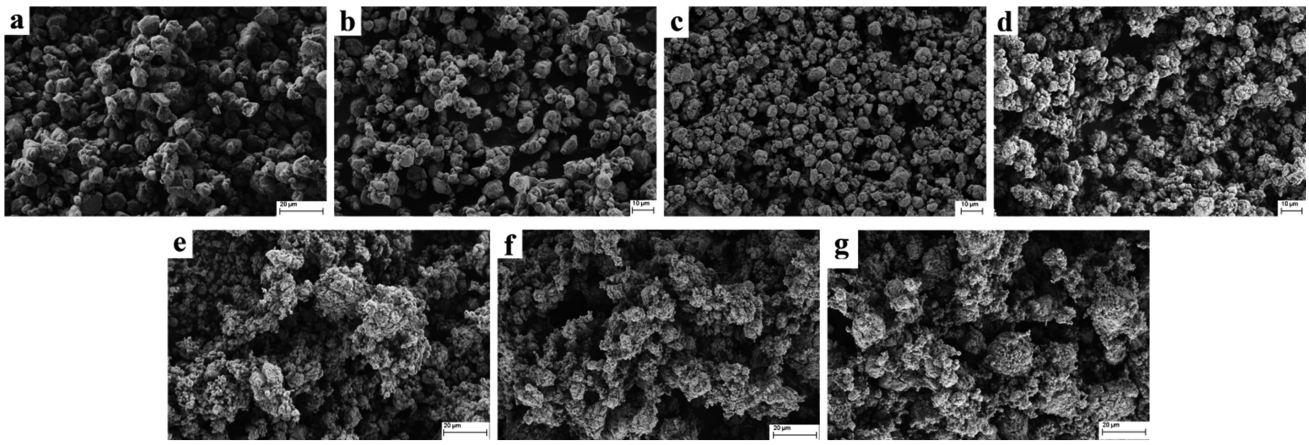


Fig. 2 SEM images for BM ZK50 alloy at **a** 3 h, **b** 5 h, **c** 10 h, **d** 19 h, **e** 31 h, **f** 40 h and **g** 45 h

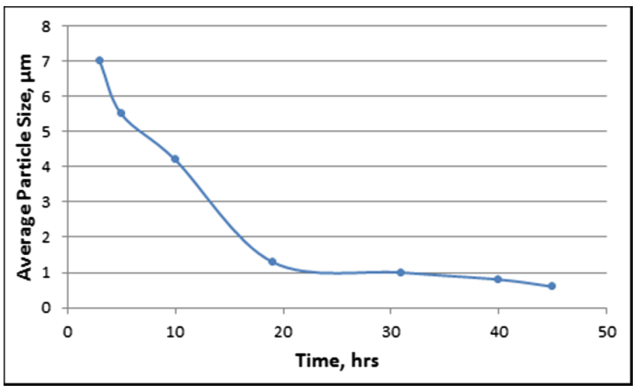


Fig. 3 Effect of milling time on the average particle size during BM

XRD patterns for the powders are shown in Fig. 4. The as-received powders as well as the BM powders were investigated in order to spot any new peaks formed by new phases from the mechanical alloying of the powders during milling. Some peaks disappeared and other formed due to the formation of the alloy. Zn peak at $2\theta = 39°$ (indicated by yellow arrow in Fig. 4) and Mg/Zr peaks at $2\theta = 47°$, 57°, 63° and 69° (indicated by black arrow in Fig. 4) started to disappear after 10 h of milling time and almost disappeared between 38 and 43 h. Moreover, the peak at $2\theta = 41°$ (indicated by red arrow in Fig. 4) was formed, which is suggested to be $MgZn_2$ by Zhou et al. [14]. Therefore, the alloying time was suggested to be 40 h under the employed BM parameters. However, Palacios-Lazcano suggested the formation of Mg-5%Zn at 60 h [9].

Broadening of the Mg peaks indicates internal structure refinement, manifested in XRD-measured crystallite size and lattice strain, as a function of milling time, as shown in Fig. 5. Significant crystallite size reduction was recorded, from 52 to 22 nm in the first 10 h of milling. This was followed by gradual reduction to 8.83 nm at 50-h. Such refinement compares favorably with the literature. For instance, Hwang et al. produced Mg powders with average crystallite size of 42 nm after 27 h of milling [7]. Grosjean et al., prepared Mg powders using BM with crystallite size of 34 nm after 40 h of milling [8]. Palacios-Lazcano reported the smallest crystallite size to date by the preparation of Mg-Zn alloy using BM, with crystallite size of 20 nm [9]. Our lattice strain measurements in Fig. 5, reflected an increasing trend with milling time, also compare closely to the work done by Anil Kumar et al. on an Mg-7.4%Al alloy [15].

Consolidated Sheets' Properties

The consolidated hot rolled thin sheets were cut into three pieces for density measurement. Each specimen was 10 mm long and 11 mm wide. The measured relative density was 128 and 125% for the conditions 1 and 2, respectively which were classified as invalid measurements. Magnesium is one of the most active elements, which can easily form oxide layer on the surface of the consolidated powders once exposed to air. Provided that the hot rolled thin sheets were about 800 and 720 in thickness for conditions 1 and 2, respectively, the oxide layer formed could have attributed to the measured high relative densities [16]. The relatively high surface area of the oxide layer-to-the sheets thickness contributed to the measured error in densities.

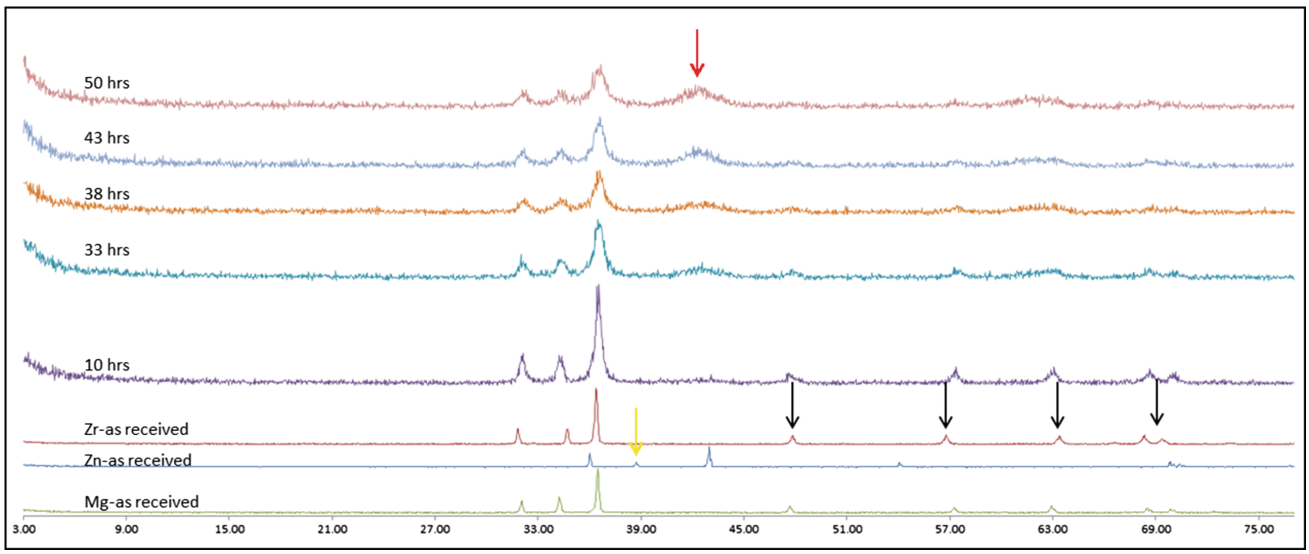


Fig. 4 XRD patterns for the as received and milled powders. Disappeared peaks are indicated by *yellow arrow* for Zn peak at $2\theta = 39°$ and *black arrows* for Mg/Zr peaks at $2\theta = 47°$, 57°, 63° and 69° while *red arrows* indicate the formed peak at $2\theta = 41°$ (Color figure online)

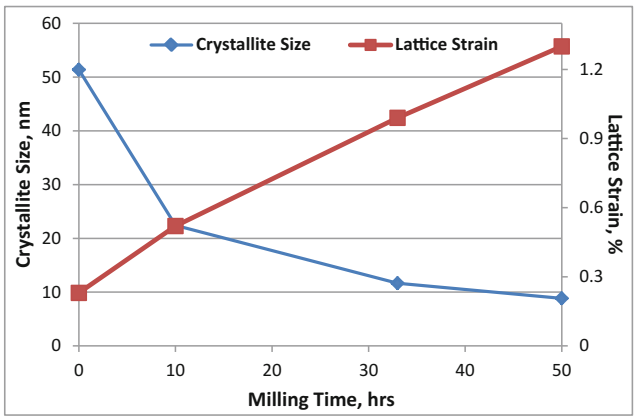


Fig. 5 The effect of milling time on the crystallite size and the lattice strain

Figure 6 shows the XRD pattern for both processing conditions. There are two types of phases are present, α-Mg indicated by the black circle in Fig. 6 and another phase indicated by the black rectangle. It was noticed the presence of the same peaks for both condition. New peaks are showed after solidification due to the formation of new phases during rolling and heating. Those peaks are indicated by red arrow. As long as this composition was not investigated before by other research groups, therefore, XRD patterns reported by different authors for ZK60 were used as a guide for interpreting the peaks. Liu et al. [17] and Gua et al. [18], reported XRD patterns for extruded ZK60 Mg alloy while Zhou et al. [14] reported XRD patterns for cast ZK60 Mg alloy. They all reported evidence for ZK alloy peaks, which was indexed as α-Mg. α-Mg is a single phase with a hexagonal closed packed crystal structure. On the other hand, the three research groups reported XRD peaks at 2θ = 24, 42 and 69 for MgZn, Mg_2Zn and $MgZn_2$ in references [17, 18] and [14] respectively.

The crystallite size of the solid sheets was calculated using Sherrer Equations giving 18.5 and 49.5 nm for condition 1 and condition 2 samples, respectively. The lattice strain energy was calculated to be 0.62 and 0.23% for each sample. It is worth noting that the condition 2 sample has coarser crystallite size because compared to condition 1. This could be attributed to the longer annealing duration of 60 min at 300 °C followed by furnace cooling for condition 2 compared to 30 min, 300 °C followed by air cooling. Accordingly, the proposed consolidation processes resulted in coarsening of the mechanically alloyed powders' internal structures by almost 2× and 4.5× for the first and second conditions, respectively.

Figures 7 and 8 show SEM images for the surfaces and cross-section of the mechanically alloyed and consolidated sheets at low and high magnifications, respectively. The investigated surfaces for conditions 1 and 2 for the processed sheets revealed the presence of needle- and flower-like structures (Fig. 7b).

According to Emley [19], when Zr content is as low as 0.4–0.5%, the grains are coarsened and Zr cores take the shape of stars of 6-platelet flower, which is clearly evident in Fig. 7b. Circles are used on the figure to indicate the 6 petalled-flowers, while arrows are pointing out to the needle structure. Moreover, dimple-like structures are observed on the sheets surface and cross-section for both processing conditions. More studies are currently undergoing to determine this kind of phase. Figure 8a, b show high magnification images for conditions 1 and 2, respectively, which demonstrates evidence for the formation of dimple-like structure both appearing on the surfaces and cross sections of the processed sheets. The average diameter of these dimples was measured to be 183 and 203 nm for condition 1 and condition 2 samples, respectively. From the XRD, the sample prepared via condition 1 has finer crystallite size and higher lattice strain,

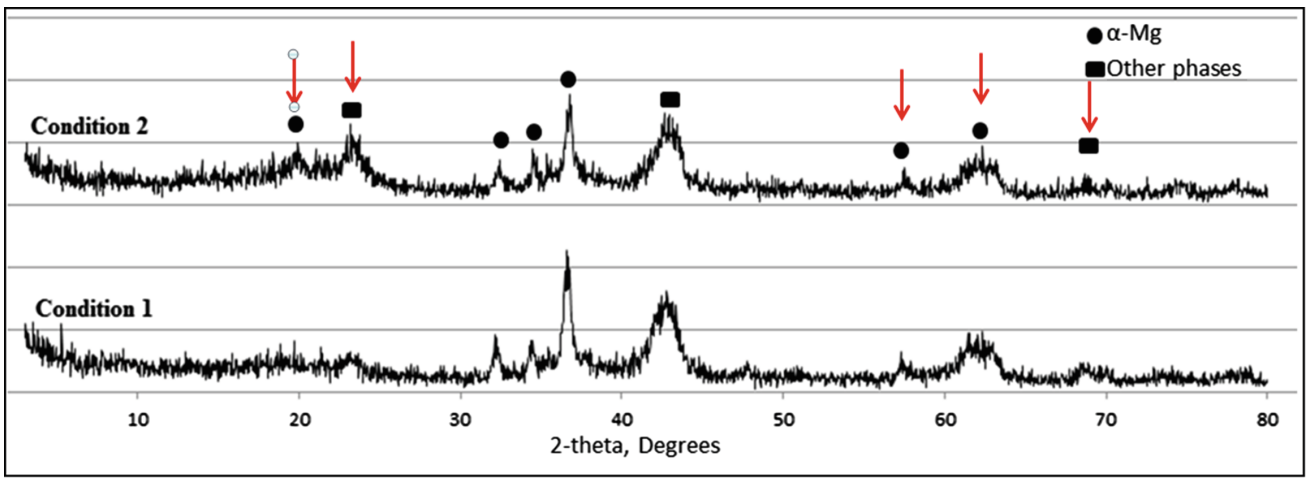


Fig. 6 XRD for the consolidated alloy sheets for conditions 1 and 2 (Color figure online)

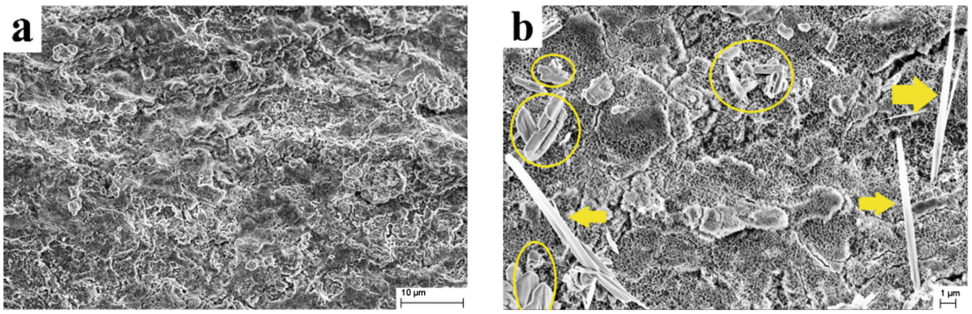


Fig. 7 SEM images for the surfaces of the consolidated sheets for **a** low and **b** high magnifications, respectively. *Yellow circles* in (**b**) indicate the 6 petalled-flowers while *arrows* are pointing out to the needle structure (Color figure online)

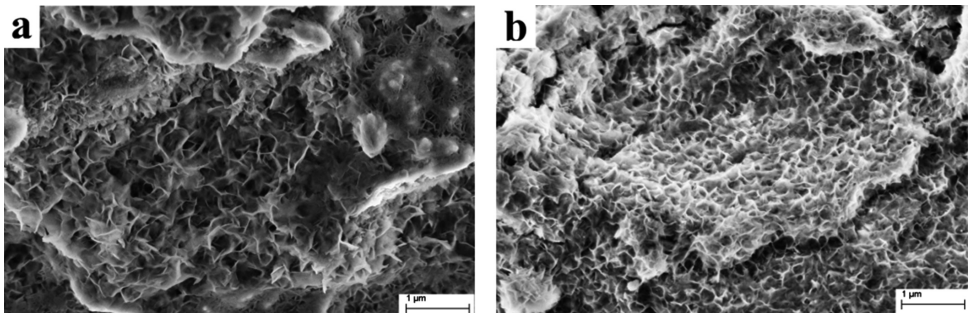


Fig. 8 SEM images for **a** Condition 1 cross-section, **b** Condition 2 surface

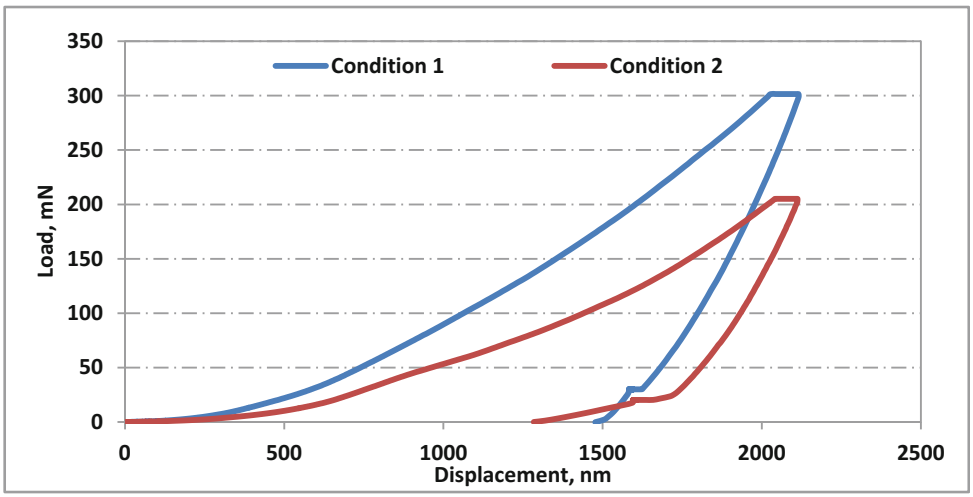


Fig. 9 Load-displacement curves for a represented test point in each condition

therefore it was expected to have higher hardness than the sample prepared at condition 1.

Figure 9 shows the Nanoindentation results displaying the load-displacement curve for a representative test points for each processing condition. It is observed that condition 1 has higher load bearing capacity compared to that of condition 2, which agrees with the measured hardness and young's modulus displayed in Fig. 10.

The values of the hardness was found to be 3 and 2.1 GP, while the bulk modulus was found to be 68 and 53.2 for conditions 1 and 2, respectively. This agrees with the calculation done from the XRD patterns for crystallite size and stored energy retained by the structures during the hot consolidation and process annealing [20]. Figure 10 shows the hardness and the young's modulus of both samples comparing to the processing conditions.

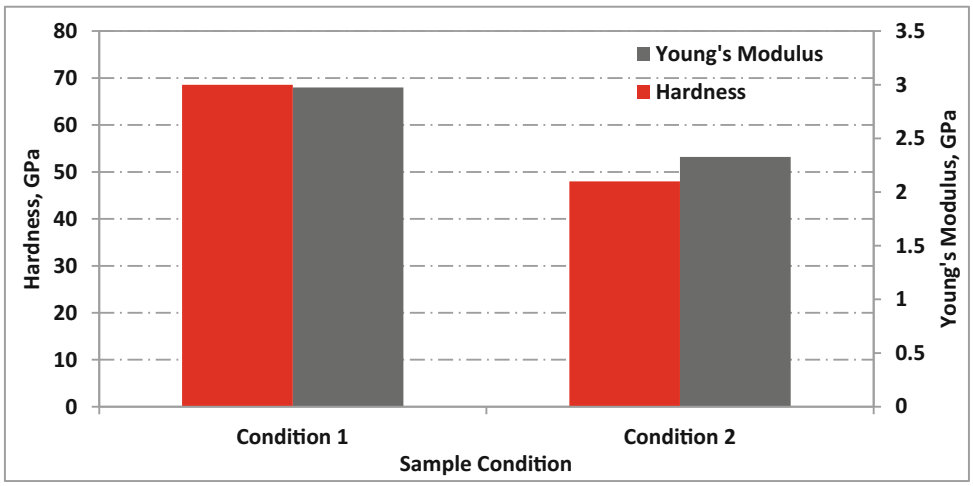


Fig. 10 Comparison between hardness and Young's Modulus for both conditions

Conclusions

ZK50 Magnesium alloy was processed from elemental powders using Mechanical milling technique. The alloy was formed at milling time ≥ 40 h of milling. The particle size and crystallite size of the mechanically milled powders were 600 nm after 45 h and 8.83 nm, respectively. The powders were successfully consolidated into thin sheets using powder rolling associated with process annealing. Powders in the bulk form processed by condition 2 show lower mechanical properties and coarser grain size due to the longer annealing time. 6-platelet flower, needle-like and dimple like structures nucleated during consolidation, which are yet under investigation. Mg ZK50 alloy bulk thin sheets fabrication via combination of MA, powder rolling and process annealing is a promising technique for the production of biocompatible and biodegradable biomedical parts.

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